

Chain Reduction as substitution: A PF-based account of anti-locality*

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1. Introduction

In current minimalist theorizing, the elimination of copies left by Internal Merge (e.g., *What_i [_{VP} broke ~~what_i]?~~*) is typically attributed to copy deletion (see, a.o., Chomsky 1995, Bobaljik 2002, Nunes 2004, Landau 2006, Bošković and Nunes 2007), formalized influentially by Nunes (2004) as the operation of Chain Reduction (1).

(1) Chain Reduction

Delete the minimal number of constituents of a nontrivial chain CH that suffices for CH to be mapped into a linear order in accordance with the LCA. [Nunes 2004:27]

In what follows, I consider an alternative hypothesis that Chain Reduction operates via substitution rather than deletion. Having formalized this idea in section 2, in section 3 I show that it opens the door for deriving complement-to-specifier anti-locality effects (ban on P-stranding) from labeling ambiguity arising in head/head structures (Chomsky 2013, 2015). The particular approach to the timing of the Labeling Algorithm that I adopt will then lead to the prediction that complement-to-specifier anti-locality violations should be resolvable through ellipsis. While a detailed discussion of this point is beyond the current scope, I touch upon it briefly in section 3.1, noting that data coming from a growing body of research on P-omission under sluicing and in fragment answers suggest that this indeed may be the case.

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2. Chain reduction as substitution

Mendes and Ranero (2021) propose that Chain Reduction can proceed via substitution, in addition to deletion. Their argument is based on adjunct extraction in K'iche' and Kaqchikel (Mayan), where the particle *wi* substitutes for lower copies under specific conditions, as illustrated in (2) and (3) (see their work for details and arguments against alternative analyses).

- (2) Jawi x-at-b'ee *(wi) iwiir?
 where COM-B2S-go *(FP) yesterday
 'Where did you go yesterday?' [K'iche', Velleman 2014 via Mendes and Ranero 2021:2]
 jawi x-at-b'ee jawi iwiir [narrow syntax]
 ↑
- (3) Ankuchi x-a-b'e (wi)?
 where COM-B2S-go (FP)
 'Where did you go?' [Patzún Kaqchikel, Mendes and Ranero 2021:2]
 ankuchi x-a-b'e ankuchi [narrow syntax]
 ↑

A similar operation, even if formalized as insertion rather than substitution, has independently been proposed by Medeiros (2021) in his discussion of Hawaiian, where the marker *ai* surfaces in movement configurations like the one in (4).

- (4) He aka ka mea a Kekoa i kū'ai ai.
 a what the thing PERS Kekoa PERF buy AI
 'What is the thing that Kekoa bought?' [Hawaiian, Medeiros 2021:168]
 [he aka] ka mea a Kekoa i kū'ai [he aka] [narrow syntax]
 ↑

The model that emerges from these studies is thus one that includes two types of Chain Reduction, one operating by deletion and the other by substitution (5).

- (5) Copy realization, XP moved [based on Mendes and Ranero 2021:2]
- a. Non-realization with **Chain Reduction as deletion**
 XP → Ø [delete XP]
 - b. Overt realization with **Chain Reduction as substitution**
 XP_[Appl(icative)] → =wi [substitute XP with /=wi/]

However, if we accept the view that some instances of Chain Reduction involve substitution, it may be worth to consider the possibility that all do, which would allow us to regain a model with a single mechanism.

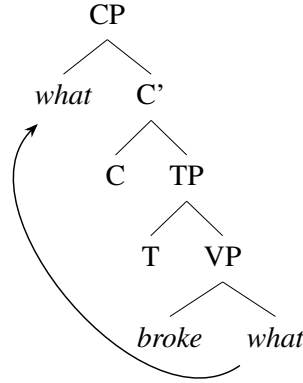
Technically, this can be achieved by substituting the set formed by merging the lower copy with the other member of the set, that is substituting the category immediately dominating the lower copy and its sister with the lower copy's sister alone (6).¹

¹Cf. Müller's (2017) Remove and Marcolli et al.'s (2025) Hopf algebraic formalization of copy deletion.

- (6) Copy non-realization with Chain Reduction as substitution, XP moved
 $\{\alpha, \{X, \dots\}\} \rightarrow \alpha$ [substitute [α XP] with XP's sister, XP a lower copy]

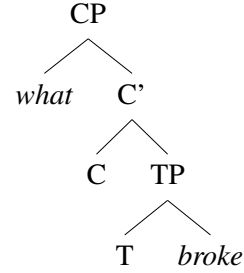
(7) *Example: What broke?*

a. Structure before Chain Reduction



b. Chain Reduction(VP) = *broke*
 [substitute [α VP] with *broke*, *what* a lower copy]

c. Structure following Chain Reduction



Such a reconceptualization of Chain Reduction as substitution enables a uniform implementation, aligned with principles of computational efficiency and Occam's razor. In addition, it paves the way for eliminating complement-to-specifier anti-locality as a stipulated syntactic constraint, as I show in the following section.

Before proceeding, I note that previous approaches to Chain Reduction largely view it as governed by a principle of deletion maximization within an economy-based perspective (a.o., Landau 2006, van Urk 2018, Bianchi 2019, Georgi and Amaechi 2023, Seguin 2025). If this type of motivation is on the right track (see Gould 2021 for a dissenting view), it can be restated for the current proposal by assuming that Chain Reduction is instead driven by representational and derivational minimization—principles familiar from the narrow syntax (Chomsky 1995, Epstein et al. 2015). Specifically, viewed as substitution, Chain Reduction minimizes the amount of structure that needs to undergo further PF computation by reducing the representation associated with the lower copy.

3. Anti-locality as labeling ambiguity

Anti-locality is concerned with the set of structural constraints that delimit the minimal domains in which syntactic operations cannot apply. Focusing on movement, it defines the minimal structure for the application of Internal Merge, and—as a result—it governs the distribution of copies. One type of anti-locality is complement-to-specifier anti-locality, which prohibits movement of a head's complement to its specifier position, illustrated in (8) (cf. Abels 2003, 2012, Grohmann 2003, Bošković 2013, 2014, Newman 2020 and references therein; see Erlewine 2016, 2020, Branen 2023, Richards forthcoming for recent approaches to Spec-to-Spec anti-locality, which I do not address here). Complement-to-specifier anti-locality accounts for the widespread ban on P-stranding observed across lan-

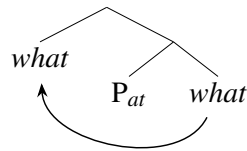
²For Chomsky (2013, 2015), adjuncts are introduced by Pair Merge rather than Set Merge and it’s unclear how labeling should be approached in this case. I put adjuncts aside here for this reason, but see Ruda submitted for a possible analysis.

- (12) Model assumed (PF side)
 narrow syntax (Merge, Agree) → cyclic Transfer → PF/SM interface (Impoverishment, Chain Reduction etc. → labeling → linearization → Vocabulary Insertion)

With these assumptions in place, I now proceed to discussing how the formalization of Chain Reduction suggested above can account for the unacceptability of P-stranding, so far attributed to complement-to-specifier anti-locality (e.g., (9) above). As I show below, the result is the same regardless of whether PPs are phases (see, e.g., Abels 2003, 2012, Bošković 2013, 2014, McInnerney 2024) or not (in line with the original conception in Chomsky 2000, 2001).

Consider the derivation of the P-stranding version of the question *What did she point at?* in (9a) above, unacceptable in Polish. First, following the assumption that PP is a phase but rejecting the anti-locality constraint means that the NP must and can move from the complement to the specifier of P (13a). When the structure reaches the relevant point at PF, by the definition in (6), Chain Reduction applies to the set containing P and NP (i.e., the traditional P'), as shown in (13b), yielding the representation in (13c).

- (13) a. Structure following Comp-to-Spec movement, PP a Spell-Out domain



- b. Chain Reduction applies to [P_{at} *what*] at PF

Chain Reduction($\{P_{at}, \textit{what}\}$) = P_{at}

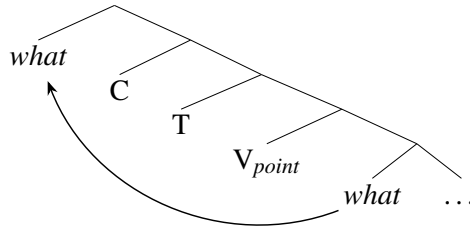
- c. Structure following Chain Reduction



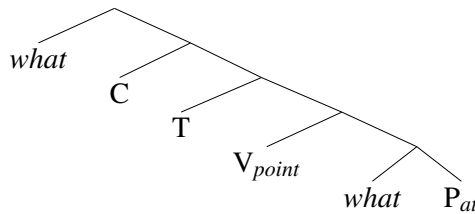
[input to Chain Reduction substituted by output]

In the narrow syntax, the *wh*-phrase undergoes further movement to Spec,C (14a) and, for explicitness, I assume that after Spell-Out the representation of the CP is integrated with the representation of the PP that has undergone Chain Reduction (14b). Because the *wh*-phrase has undergone the second movement, Chain Reduction needs to apply to the PP again (14c). This results in the V/P head/head structure in (14d), which—in accordance with the model adopted here (see (11) and (12) above)—cannot be labeled and cannot therefore be linearized (Takita 2020).

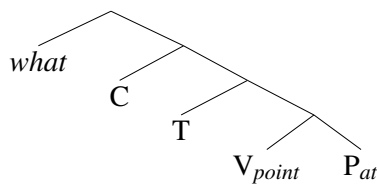
- (14) a. Structure following movement of *what* to Spec,C, CP a Spell-Out domain [narrow syntax]



- b. Structure following integration with PP at PF
[[*what* C [*she pointed*]] integrated with the reduced PP]

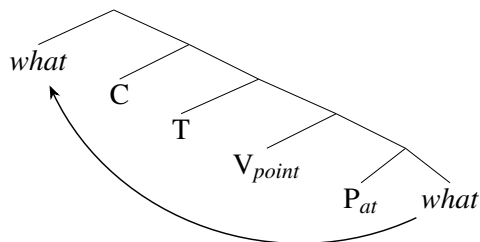


- c. Chain Reduction($\{ \textit{what}, P_{at} \} \}) = P_{at}$ [Chain Reduction applies to [*what* P_{at}] at PF]
d. Structure following Chain Reduction



On the alternative view that PPs are not phasal, the *wh*-phrase moves directly from the complement of P to Spec,C (15a) and Chain Reduction applies to the PP only once (15b). The result is still the same, though, since substituting the PP set (i.e., simplifying, $\{P, \{wh\text{-}NP\}\}$) with the *wh*-NP's sister, as required by (6), yields just P (15c). Hence, regardless of whether PPs are phases or not, P-stranding results in a representation that cannot be labeled and, what follows, cannot undergo linearization that is needed for externalization by the sensorymotor system.

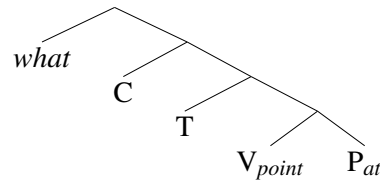
- (15) a. Structure following movement of *what* to Spec,C, CP a Spell-Out domain [narrow syntax]



- b. Chain Reduction($\{ P_{at}, \textit{what} \} \}) = P_{at}$ [Chain Reduction applies to [P_{at} *what*] at PF]

c. Structure following Chain Reduction

[input to Chain Reduction substituted by output]



In sum, no narrow syntactic constraints need to be postulated to block the movement of *wh*-NPs from the complement of P to Spec,P to account for the ban on P-stranding. Its unacceptability can be derived on the assumption that Chain Reduction involves set substitution, combined with shifting the timing of Chomsky's Labeling Algorithm to PF, as argued for on independent grounds in Takita 2020 and Ruda 2025. Interestingly, both Takita (2020) and Ruda (2025) discuss cases where deletion can rescue unlabelable and unlinearizable derivations, and this seems to be the case with P-stranding as well, at least in some contexts.

3.1 P-omission in clausal ellipsis

Non-P-stranding languages have been observed to permit P-omission under sluicing and in fragment answers, at least in some contexts (a.o., Almeida and Yoshida 2007, Szczegielniak 2008, 2016, Rodrigues et al. 2009, Caha 2011, van Craenenbroeck 2011, Nykiel 2013, Abels 2017, Molimpakis 2019, Ionova 2020, Griffiths et al. 2021, Nykiel and Kim 2022, Stigliano 2022, Hassen and Abeillé 2025; contra Merchant 1999; this also holds for English in the contexts where P-stranding is disallowed (Ross 1967, 1969, Lasnik 2005)). While these observations come from studies employing different methods, grounded in different theoretical frameworks, and drawing different types of analytical conclusions, it is rather undeniable at this point that the patterns leading to Merchant's (1999) Form-Identity Generalization II (a.k.a. P-Stranding Generalization, quoted here in (16)) are not as clear as originally believed.³

- (16) *Form-Identity Generalization II*: A language L will allow preposition stranding under sluicing iff L allows preposition stranding under regular *wh*-movement.
Languages: Greek, German, Dutch, Russian, Polish, Bulgarian, Serbo-Croatian, Slovene, Persian, Catalan, Spanish, French, Italian, Hebrew, and Moroccan Arabic and Basque
 [Merchant 1999:92]

For example, adding to the data provided in the research cited above, (17)–(18) illustrate that the National Corpus of Polish (Przepiórkowski et al. 2012) contains sluicing structures and fragment answers with P omission.⁴

³The P-Stranding Generalization should be contextualized within the standard analysis of sluicing as involving the movement of the remnant from within the ellipsis site, followed by clausal deletion. I adopt this analysis here, though some of the authors cited above advocate alternative approaches.

⁴Thanks for help with extracting the data go to Karolina Zaremba and Wiktor Eżlakowski.

- (17) Ja w Poznaniu kupiłem za 27.90 w **jakiejś** **księgarni**
 I in Poznań bought.1SG for 27.90 in some.LOC bookstore.LOC
językowej, ale nie pamiętam **jakiej.**
 language.ADJ.LOC but not remember.1SG which.LOC
 ‘I bought it in Poznań for 27.90 in some language bookstore, but I don’t remember
 which (one).’ [National Corpus of Polish (Przepiórkowski et al. 2012)]
 ... but I don’t remember *[[which] I bought {PP [which] [-in [which] }]]*.

- (18) A: ... ten niepozbianiec F. I H. H. sobie zemdlął w szkole.
 that scatterbrain F. I H. H. himself faint.PST.3SG in school
 ‘... that scatterbrain F. I H. H. fainted at school.’
 B: O, naprawdę, czy żartował? A: Naprawdę.
 oh really or joke.PST.3SG really
 ‘Oh, really or was he joking?’ ‘Really.’
 B: **Na jakiej** **lekcji?** A: **Fizyce.**
 in which.LOC lesson.LOC physics.LOC
 ‘During which lesson?’ ‘Physics.’
 [National Corpus of Polish (Przepiórkowski et al. 2012)]
[[physics] he fainted {PP [physics] [-during [physics] }]].

Similarly, even though Brazilian Portuguese typically requires Ps to pied-pipe with the *wh*-NP they introduce (19), Corpus do Português (Davis 2016–) likewise includes sluiced clauses and fragment answers with P-omission ((20)–(21)).⁵

- (19) *(Com) quem ele dançou (*com)?
 with who he danced with
 ‘Who did he dance with?’ [Brazilian Portuguese]
- (20) E aí, é viciada **em alguma série médica? Qual?**
 and then be.PRS.3SG addicted to some series medical which
 ‘So, are you addicted to any medical series? Which one?’
 [Corpus do Português (Davis 2016–)]
[[which] you addicted {PP [which] [-to [which] }]]?
- (21) A: **Com que roupa** está agora?
 with what clothing be.PRS.3SG now
 ‘What are you wearing now?’

Note that the locative case marking of the remnant, required by the P, excludes the possibility of a cleft-based derivation of these examples.

⁵Thanks for help with extracting the data go to Karolina Zaremba and Samuel Figueira Cardoso.

B: **Shorts e regata.**

shorts and tank.top

‘Shorts and a tank top.’

[Corpus do Português (Davis 2016–)]

[[*shorts and tank top*] you are _{PP} [*sh. and t. t.*] [*with* [*sh. and t. t.*]]]].

While patterns of this type require further empirical and theoretical research, it is clear that if anti-locality is a narrow syntactic constraint (or if the effects attributed to it are excluded in alternative, narrow syntactic ways, as in Adger’s 2025 mereological framework), P-stranding is expected to be impossible regardless of ellipsis. On the other hand, if the effects of anti-locality arise at PF, ultimately as a result of a labeling and, what follows, linearization failure, ellipsis can potentially rescue the derivation (Fox and Pesetsky 2005, Takita 2020, Mendes and Kandybowicz 2023, Ruda 2025), supporting the PF-based approach. Leaving a more detailed discussion of this point for future work, I end this section by noting that neither Dynamic Antisymmetry (Moro 2000) nor Labeling Theory (Chomsky 2013, 2015) alone can account for the ban on P-stranding and eliminate anti-locality from the grammar (contra Ott 2015; see also Blümel 2017). This is because on both of these approaches lower copies are ignored, and therefore the *wh*-NP in Spec,P is predicted to be unproblematic, following its movement to Spec,C.

In sum, when the formalization of Chain Reduction proposed in (6) is combined with the assumption that labeling takes place at PF, the effects of complement-to-specifier anti-locality can be derived, providing an account of the ban on P-stranding (where observed; see Ruda submitted for an analysis of languages like English, where P-stranding is available). The number of stipulated syntactic can thus be reduced, simultaneously paving the way toward explaining how and why P-stranding violations are rescued by ellipsis (see Kalyakin and Kasenov forthcoming for a related discussion of Russian, though with an analysis that targets only (hypothesized) Left Branch Extractions).

Since complete copy elimination is not the only pattern of lower copy realization observed cross-linguistically, in the following section I offer a fuller picture of Chain Reduction as substitution.

4. Extension: Overt realizations of chain links

Beyond substitution with an independent particle (see (2)–(4) above), other overt realizations of chain links are multiple verb copy spell-out and pronoun copying (see, a.o., Nunes 2004, Landau 2006, Trinh 2009, van Urk 2018, Georgi and Amaechi 2023, and references therein). Multiple verb copy spell-out (22), which is cross-linguistically widespread (a.m.o., Abels 2001, Landau 2006 and references therein) and pronoun copying involving a full realization of a lower copy ((23); see, e.g., Bruening 2006, van Urk 2018), arise when Chain Reduction does not eliminate the/a lower copy.

- (22) **Čitat’** (–to) Ivan eë **čitaet**, no ničego ne ponimaet.
 read.INF (TO) Ivan her reads but nothing not understands
 ‘Ivan does read it, but he doesn’t understand a thing.’

[Russian, Abels 2001:1]

- (23) **Tayuwe** kt-itom-ups [_{CP} **tayuwe** apc kt-ol-i malsanikuwam-ok]?
 when 2-say-DUB when again 2-thus-go store-LOC
 ‘When did you say you’re going to go to the store?’
 [Passamaquoddy, Bruening 2006:26 via van Urk 2018:939]

However, pronoun copying can also instantiate a partial realization of a chain link when a copy of a lexical NP is realized with a pronoun (24). This context can also be analyzed as substitution, in particular substitution of a copy with its subpart, for example as in (25), which restates the deletion-based account for (24).

- (24) **Bagi-zi** Musa gàn [_{CP} gànán **a:** nì enyà] o.
 man-PL Musa say C 3PL beat drum FOC
 ‘Musa said that THE MEN beat a drum.’
 [Nupe, Kandybowicz 2007:124 via van Urk 2018:939]

- (25) [_{NumP} Num [_{nP} n $\sqrt{\quad}$]] $\rightarrow$ Num [restatement of the deletion-based account]

Partial copy realization of this type can thus be formalized as in (26). Importantly, insights on the constraints and conditions on its application carry over from the deletion-based accounts cited above.

- (26) Partial copy realization with Chain Reduction as substitution, XP (i.e., $\{\alpha, \beta\}$)
 moved
 $\{\alpha, \beta\} \rightarrow \alpha$ [substitute XP with its subpart]

5. Summary and conclusion

Starting with the question of how the grammar handles the PF side of chains, I have offered an alternative to the deletion-based account in terms of set substitution. Previous detailed studies of copy realization in various languages and contexts cited above suggest the following substitution options:

- (27) a. Copy non-realization
 $\{\alpha, \{X, \dots\}\} \rightarrow \alpha$ [substitute [α XP] with XP’s sister, XP a copy]
 b. Partial copy realization
 $\{\alpha, \beta\} \rightarrow \alpha$ [substitute XP (i.e., $\{\alpha, \beta\}$) with its subpart, XP a copy]
 c. Alternative copy realization
 $\{X, \dots\} \rightarrow \alpha$ [substitute XP with α , α independent of XP, XP a copy]

On the view that Vocabulary Insertion is the replacement of a Q-variable in a (terminal) head by a morphological exponent (Halle 1990, Embick 2010, 2015), (27c) can further be understood as an extreme instantiation of (27b), in that the XP is in this case substituted with a Q-variable included in one of its terminals (28). The number of substitution options is thus reduced to two.

- (28) Alternative copy realization as partial copy realization
 $\{\alpha_{[Q]}, \beta_{[Q]}\} \rightarrow Q$ [substitute XP (i.e., $\{\alpha, \beta\}$) with a Q variable in XP, XP a copy]

All in all, substitution emerges as an operation available at PF beyond Vocabulary Insertion, for which it has previously been proposed (Halle 1990, Embick 2015) and perhaps also Impoverishment, for which it likewise may be needed (cf. Noyer 1998).

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